SKILL-1

ID No: 2400030040

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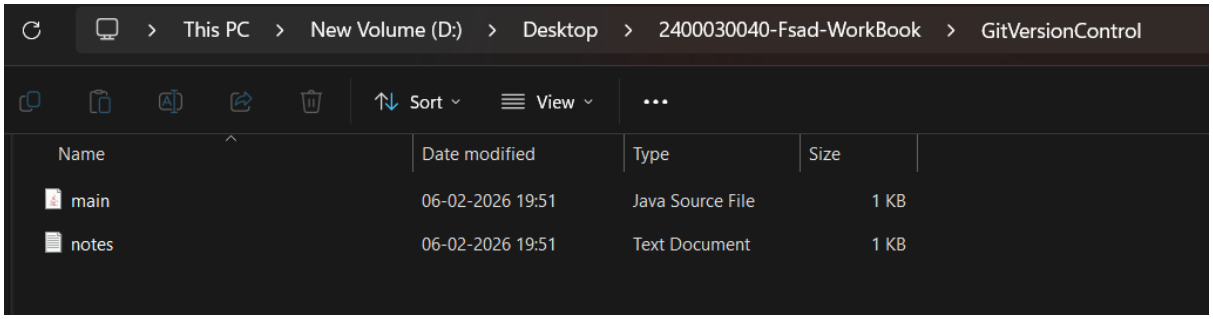
Git Version Control

Step 1: Create a new project folder and initialize Git (git init).

```
D:\Desktop\2400030040-Fsad-WorkBook\GitVersionControl>cd ..

D:\Desktop\2400030040-Fsad-WorkBook>git init
Reinitialized existing Git repository in D:/Desktop/2400030040-Fsad-WorkBook/.git/

D:\Desktop\2400030040-Fsad-WorkBook>git config --global user.name "Aravind2k6"

D:\Desktop\2400030040-Fsad-WorkBook>git config --global user.email "nelavelliaravind@gmail.com"

D:\Desktop\2400030040-Fsad-WorkBook>
```

Step 2: Create files (main.java, notes.txt), check status, and add to staging.

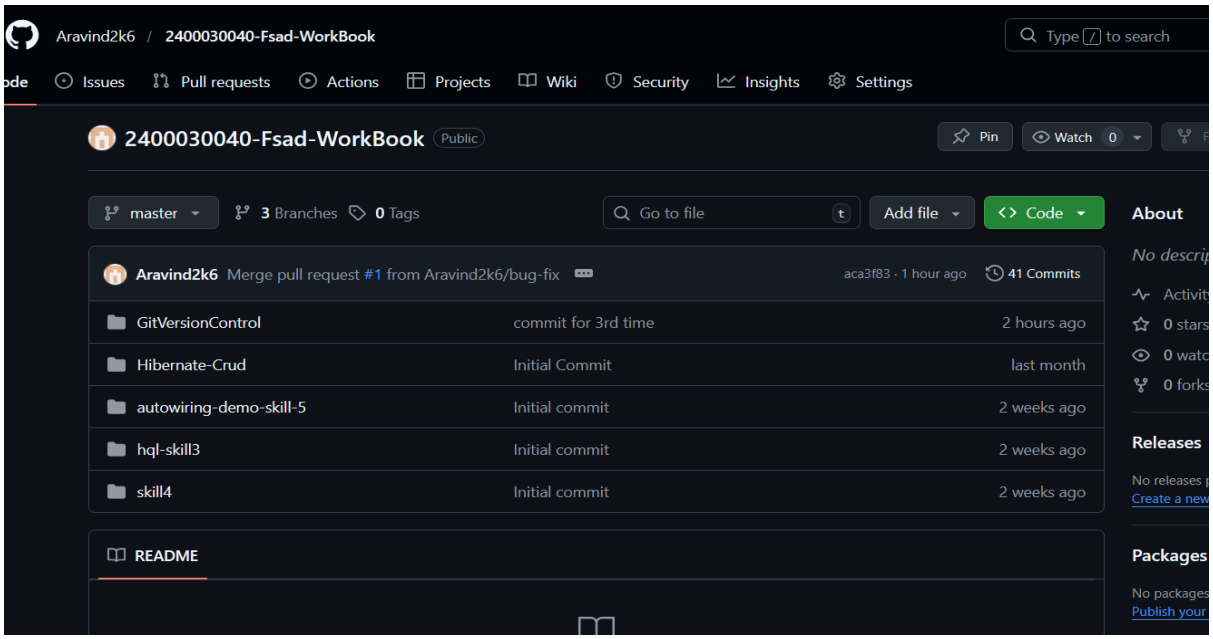



```
D:\Desktop\2400030040-Fsad-WorkBook>git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Experiment1.zip
        GitVersionControl/
        Hibernate-Crud.zip
        autowiring-demo-skill-5.zip
        hql-skill3.zip
        main.java
        notes.txt
        skill4.zip

nothing added to commit but untracked files present (use "git add" to track)
```

Step 3: Commit the staged files with a meaningful message.

```
D:\Desktop\2400030040-Fsad-WorkBook>git add .

D:\Desktop\2400030040-Fsad-WorkBook>git commit -m "Initial commit"
[master ac2dd1b] Initial commit
 9 files changed, 7 insertions(+)
 create mode 100644 Experiment1.zip
 create mode 100644 GitVersionControl/main.java
 create mode 100644 GitVersionControl/notes.txt
 create mode 100644 Hibernate-Crud.zip
 create mode 100644 autowiring-demo-skill-5.zip
 create mode 100644 hql-skill3.zip
 create mode 100644 main.java
 create mode 100644 notes.txt
 create mode 100644 skill4.zip
```

Step 4: Create a GitHub repository, connect remote URL, and push changes



Step 5: Create first branch (feature-update), modify files, add and commit.

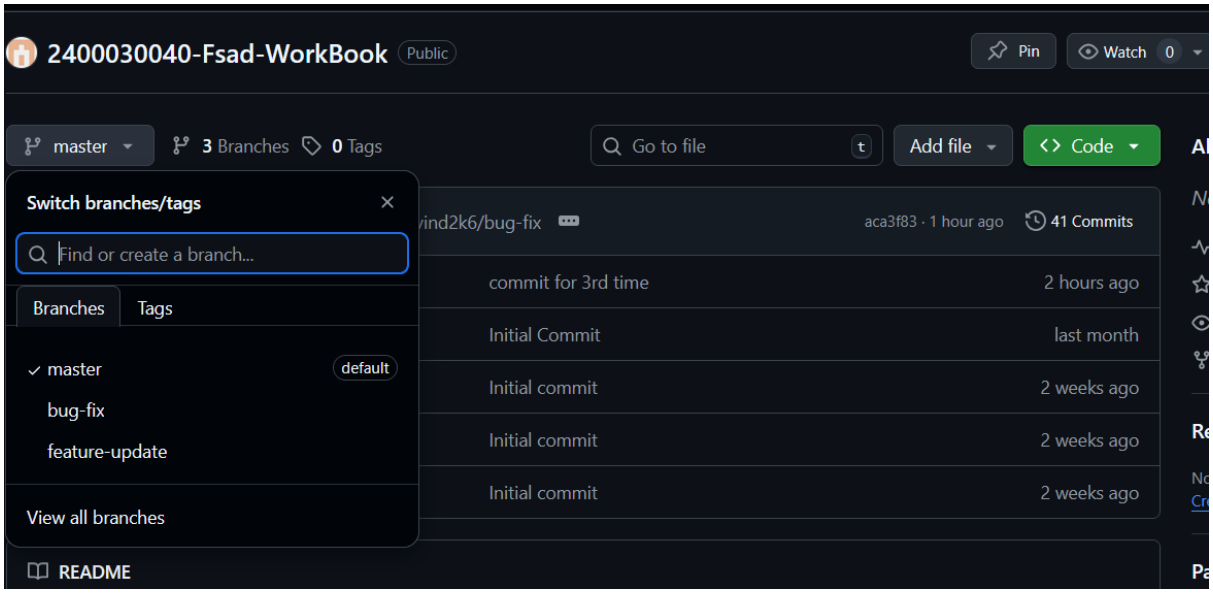
```
D:\Desktop\2400030040-Fsad-WorkBook>git checkout -b feature-update
Switched to a new branch 'feature-update'

D:\Desktop\2400030040-Fsad-WorkBook>git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'?
hint: Disable this message with "git config set advice.addEmptyPathspec false"

D:\Desktop\2400030040-Fsad-WorkBook>git add GitVersionControl

D:\Desktop\2400030040-Fsad-WorkBook>git commit -m "Updated feature in notes file"
[feature-update 4134318] Updated feature in notes file
 1 file changed, 1 insertion(+)

D:\Desktop\2400030040-Fsad-WorkBook>git checkout main
error: pathspec 'main' did not match any file(s) known to git
```

Step 6: Create second branch (bug-fix), apply changes, add and commit.

```
D:\Desktop\2400030040-Fsad-WorkBook>git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

D:\Desktop\2400030040-Fsad-WorkBook>git checkout -b bug-fix
Switched to a new branch 'bug-fix'

D:\Desktop\2400030040-Fsad-WorkBook>git add GitVersionControl

D:\Desktop\2400030040-Fsad-WorkBook>git commit -m "Bug fix changes"
On branch bug-fix
nothing to commit, working tree clean

D:\Desktop\2400030040-Fsad-WorkBook>git add GitVersionControl

D:\Desktop\2400030040-Fsad-WorkBook>git commit -m "Bug fix changes"
[bug-fix 1036c47] Bug fix changes
 1 file changed, 1 insertion(+)
```

Step 7: Switch to main branch and merge both branches.

```
D:\Desktop\2400030040-Fsad-WorkBook>git branch
  bug-fix
* feature-update
  master
```



Step 8: Resolve merge conflicts if any and complete merge.

Git Repository:

https://github.com/Aravind2k6/2400030040-Fsad-WorkBook.git

VIVA QUESTIONS:

1.Why is Git used in development?

A. Git is used in software development to manage and track changes made to source code over time. It allows developers to work on the same project simultaneously without interfering with each other's work. Git maintains a complete history of code changes, making it easy to identify who made a change and when. It supports branching, which helps developers work on new features or fix bugs separately. Git also helps in restoring previous versions of the project if errors occur.

2.What information is stored in Git configuration?

A. Git configuration stores user-specific and system-related settings required for Git operations. It includes the developer's name and email address, which are attached to every commit. It also stores default editor settings used for commit messages. Configuration may include merge and pull preferences and shortcut aliases for Git commands. These settings help Git function properly and maintain commit ownership information.

3.What is Maven and why is it used?

A. Maven is a build automation and project management tool mainly used for Java-based applications. It simplifies the build process by automatically compiling source code, running tests, and packaging applications. Maven manages project dependencies by downloading required libraries automatically from repositories. It provides a standard project structure, which makes projects easier to understand and maintain. Maven reduces manual effort and improves consistency across different development environments.

4.What is the role of the POM.xml file?

A. The POM.xml file is the core configuration file of a Maven project. It contains essential project information such as project name, version, and group details. The file defines all dependencies required by the project and specifies build plugins and configurations. It controls how the project is built, tested, and packaged. Without the POM.xml file, Maven cannot manage or build the project.

5.Explain Maven Build Life Cycle.

A. The Maven build life cycle is a predefined sequence of phases used to build and manage a project. It starts with validating the project structure, followed by compiling the source code. After compilation, tests are executed to verify correctness. The project is then packaged into a deployable format such as JAR or WAR. Finally, the build is installed in the local repository or deployed to a remote repository, ensuring a structured and automated build process.